

DPP No. # 1 (JEE-ADVANCED)

Total Marks : 33

Max. Time : 30 min.

Single choice Objective Questions ('-1' negative marking) Q. 1 to Q.3

(3 marks 3 min.)

[09, 09]

Multiple choice Objective Questions ('-1' negative marking) Q.4 to Q.8

(4 marks 3 min.)

[20, 15]

Question No.	1	2	3	4	5	6	7	8	Total
Marks Obtained									

- Let A and B two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is
(A) 256 (B) 220 (C) 219 (D) 211
- If $P = (\tan(3^{n+1}\theta) - \tan \theta)$ and $Q = \sum_{r=0}^n \frac{\sin(3^r \theta)}{\cos(3^{r+1} \theta)}$, then
(A) $P = 2Q$ (B) $P = 3Q$ (C) $2P = Q$ (D) $3P = Q$
- Let $[x]$ denote the greatest integer part of a real number x , if
$$M = \sum_{n=1}^{40} \left[\frac{n^2}{2} \right]$$
 then M equals
(A) 5700 (B) 5720 (C) 5740 (D) 11060
- The equation $(3 + \cos x)^2 = 4 - 2 \sin^8 x$ has
(A) exactly one solution in $x \in (0, 3\pi)$ (B) exactly three solutions
(C) exactly two solutions $x \in [0, 5\pi)$ (D) infinite solutions
- If quadratic equation $x^2 + 2(a + 2b)x + (2a + b - 1) = 0$ has unequal real roots for all $b \in \mathbb{R}$ then the possible values of a can be equal to
(A) 5 (B) -1 (C) -10 (D) 3
- If $\frac{a+b}{x} = \frac{b+c}{y} = \frac{c+a}{z}$, then
(A) $\frac{a+b+c}{x+y+z} = \frac{ab+bc+ca}{ay+bz+cx}$ (B) $\frac{c}{y+z-x} = \frac{a}{x+z-y} = \frac{b}{x+y-z}$
(C) $\frac{c}{x} = \frac{a}{y} = \frac{b}{z}$ (D) $\frac{a+b}{b+c} = \frac{a+b+x}{b+c+y}$
- Consider the function $f(x) = mx^2 + (2m - 1)x + (m - 2)$
(A) If $f(x) > 0 \forall x \in \mathbb{R}$ then $m \in \left(-\infty, -\frac{1}{4}\right)$
(B) The number of integral values of 'm' greater than -5 so that $f(x) < 0 \forall x \in \mathbb{R}$ are 4
(C) The number of integral values of m less than 50 so that the roots of the quadratic equation $f(x) = 0$ are rational, are 6
(D) The curve $y = f(x)$ touches the X-axis for $m = -\frac{1}{4}$
- If a, b, c are natural number satisfying $a^2 + b^2 = c^2$, then :
(A) at least one of a, b, c is even. (B) at most two of a, b, c can be odd.
(C) abc is divisible by 30. (D) abc is divisible by 8.

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1. (C) 2. (A) 3. (D) 4. (ACD) 5. (BC) 6. (ABD)
7. (BCD) 8. (AB)

Solution
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1. $n(A) = 2$

$n(B) = 4$

$n(A \times B) = 8$

$${}^8C_3 + {}^8C_4 + \dots + {}^8C_8 = 2^8 - {}^8C_0 - {}^8C_1 - {}^8C_2$$

$$= 256 - 1 - 8 - 28$$

$$= 219$$

2.
$$\frac{\sin \theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \dots + \frac{\sin 3^n \theta}{\cos 3^{n+1} \theta}$$

$$= \frac{1}{2} \left[\frac{2 \sin \theta}{\cos 3\theta} + \frac{2 \sin 3\theta}{\cos 9\theta} + \dots + \frac{2 \sin 3^n \theta}{\cos 3^{n+1} \theta} \right]$$

$$= \frac{1}{2} [\tan 3\theta - \tan \theta + \tan 3^2 \theta - \tan 3\theta + \dots + \tan 3^{n+1} \theta - \tan 3^n \theta]$$

$$= \frac{1}{2} [\tan 3^{n+1} \theta - \tan \theta] = \frac{P}{2} \Rightarrow P = 2Q$$

3. $[x] = x - \{x\}$

$$\sum_{n=1}^{40} \frac{n^2}{2} - \sum_{n=1}^{40} \left\{ \frac{n^2}{2} \right\}$$

$$\frac{1}{2} \cdot \frac{n(n+1)(2n+1)}{6} - \left[\left\{ \frac{1}{2} \right\} + \left\{ \frac{4}{2} \right\} + \dots + \left\{ \frac{1000}{2} \right\} \right]$$

$$\frac{1}{2} \cdot \frac{40 \times 41 \times 81}{6} - \left[\left(\frac{1}{2} \right) \times 20 \right]$$

$$11070 - 10 = 11060$$

4. $4 \leq \text{LHS} \leq 16$ and $2 \leq \text{RHS} \leq 4 \Rightarrow \text{LHS} = \text{RHS} = 4 \Rightarrow \cos x = -1 \Rightarrow x = (2n+1)\pi \Rightarrow x = \pi, 3\pi$

5. $x^2 + 2(a+2b)x + (2a+b-1) = 0$
 $D > 0 \Rightarrow 4(a+2b)^2 - 4(2a+b-1) > 0$
 $a^2 + 4b^2 + 4ab - 8a - 4b + 4 > 0$
 $4b^2 + b(4a-4) + (a^2 - 8a + 4) > 0 \dots (ii)$
 $b \in \mathbb{R}$ so if equation (ii) is quadratic in b then it's $D < 0$
 $b \in \mathbb{R} \quad (ii) \quad b \quad D < 0$
 $16(a-1)^2 - 16(a^2 - 8a + 4) < 0$
 $a^2 - 2a + 1 - a^2 + 8a - 4 < 0$
 $6a - 3 < 0$
 $a < 1/2$

6.
$$\frac{a+b}{x} = \frac{b+c}{y} = \frac{c+a}{z} = \frac{2(a+b+c)}{x+y+z} \dots (i)$$

$$\frac{ca+cb}{cx} = \frac{ab+ac}{ay} = \frac{bc+ab}{bz} = \frac{2(ab+bc+ca)}{ay+bz+cx} \dots\dots\dots (ii)$$

from (i)& (ii)

$$(B) \quad \frac{a+b}{x} = \frac{b+c}{y} = \frac{c+a}{z}$$

(ii) + (iii) - (i)

$$\frac{2c}{y+z-x} = \frac{2a}{x+z-y} = \frac{2b}{x+y-z}$$

$$(D) \quad \frac{x}{a+b} = \frac{y}{b+c} = \frac{z}{a+c}$$

$$\frac{x+a+b}{a+b} = \frac{y+b+c}{b+c}$$

$$\frac{a+b}{b+c} = \frac{a+b+x}{b+c+y}$$

7. (A) For $m = 0$ $f(x) \nless 0 \forall x \in \mathbb{R}$

$$m > 0, \quad D < 0$$

$$\Rightarrow (2m-1)^2 - 4m(m-2) < 0$$

$$4m^2 - 4m + 1 - 4m^2 + 8m < 0$$

$$m < -\frac{1}{4} \quad \therefore m \in \phi$$

(B) $f(x) < 0 \forall x \in \mathbb{R}$ When $m < 0$ & $D < 0 \Rightarrow m \in \left(-\infty, -\frac{1}{4}\right)$

$\therefore$ Values of m greater than -5 are

$\therefore m = -5, -4, -3, -2, -1$

Number of values of $m = 4$

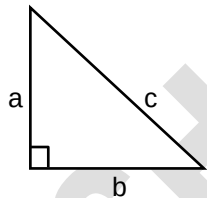
$$m = 4$$

(C) $D = 4m + 1 = \text{square of a rational number}$

$$\text{If } m = 2, 6, 12, 30, 42$$

Number of values = 6

8.



$$a^2 + b^2 = c^2$$

if $a \in \text{odd (except 1)}$

$$b = \frac{a^2 - 1}{2} \quad ; \quad c = \frac{a^2 + 1}{2}$$

so at most two of a, b, c can be odd

so atleast one of a, b, c can be even a, b, c is divisible by 3 $\rightarrow \frac{3^2}{2} \rightarrow 4, 5$

$$5 \rightarrow \frac{25}{2} \rightarrow 4, 5$$

$$a^2 + b^2 = c^2$$

$$\text{if } a \in \text{odd } b = \frac{a^2 - 1}{2} \quad c = \frac{a^2 + 1}{2}$$

so, atmost 2 a, b, c can be odd & atleast are of a, b, c is even